EXPERIMENT :-4(B)

CODE:-

```
#include <stdio.h>

#define INF 9999

int main() {
   int n, i, j, count;
   int graph[20][20];
   int selected[20] = {0};
   int x, y, min, cost = 0;

   printf("Enter number of vertices: ");
   scanf("%d", &n);

   printf("Enter adjacency matrix:\n");

   for (i = 0; i < n; i++)
      for (j = 0; j < n; j++)
         scanf("%d", &graph[i][j]);

   selected[0] = 1;

   printf("\nEdges in MST:\n");

   for (count = 0; count < n - 1; count++) {
      min = INF;
```

```
    for (i = 0; i < n; i++) {
      if (selected[i]) {
        for (j = 0; j < n; j++) {
          if (!selected[j] && graph[i][j] < min) {
            min = graph[i][j];
            x = i;
            y = j;
          }
        }
      }
    }

    printf("%d - %d = %d\n", x, y, min);
    cost += min;
    selected[y] = 1;
  }

  printf("Minimum Cost = %d\n", cost);

  return 0;
}
```

Example adjacency matrix

```
0 10 6 5
10 0 0 15
6 0 0 4
5 15 4 0
```

Output:

```
Edges in MST:
0 - 3 = 5
3 - 2 = 4
0 - 1 = 10

Minimum Cost = 19
```